

UNIVERSIDAD AUTONOMA DE NUEVO LEON



Facultad de Ciencias Físico Matemáticas

(FCFM)

Unidad de Aprendizaje:

Desarrollo web:

Tecnologías servidor

Unidad 03

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Matricula: 2069855

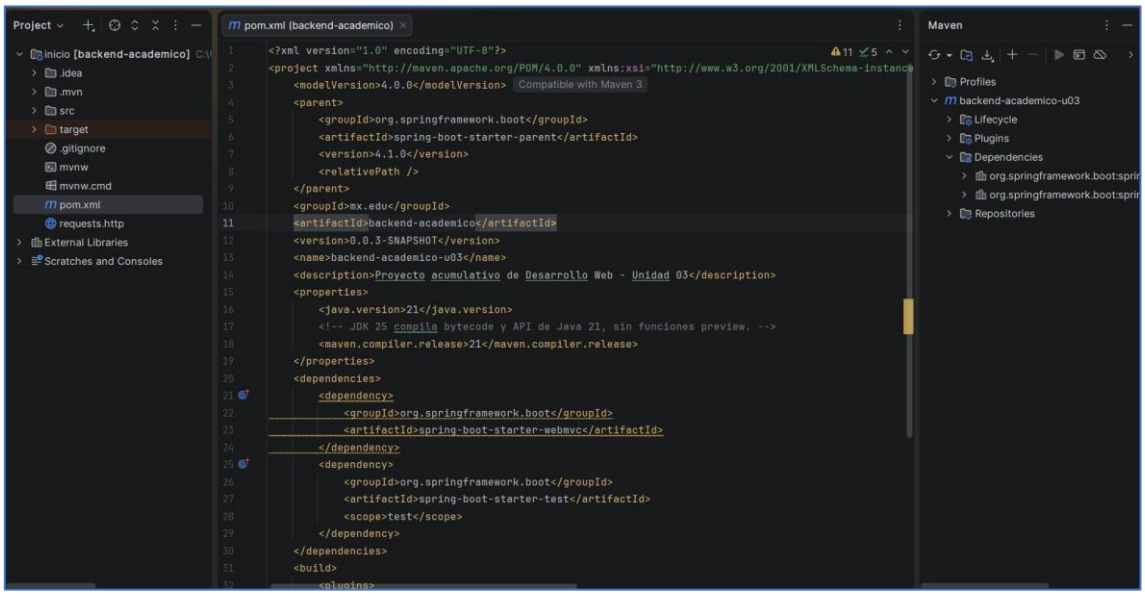
Grupo: 031

Profesor: Gustavo Adolfo Rodríguez Martínez

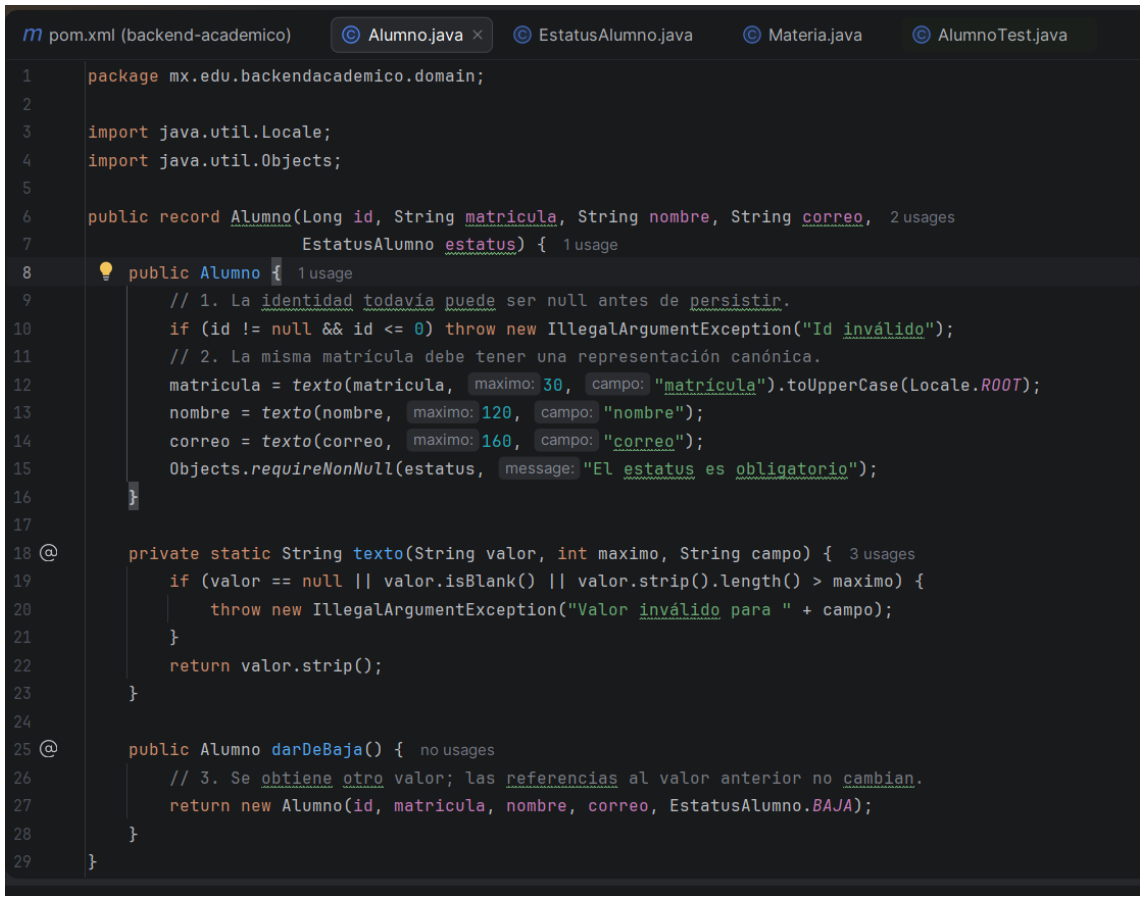
Carrera: LCC

Monterrey N.L. a; 09 de octubre de 2026

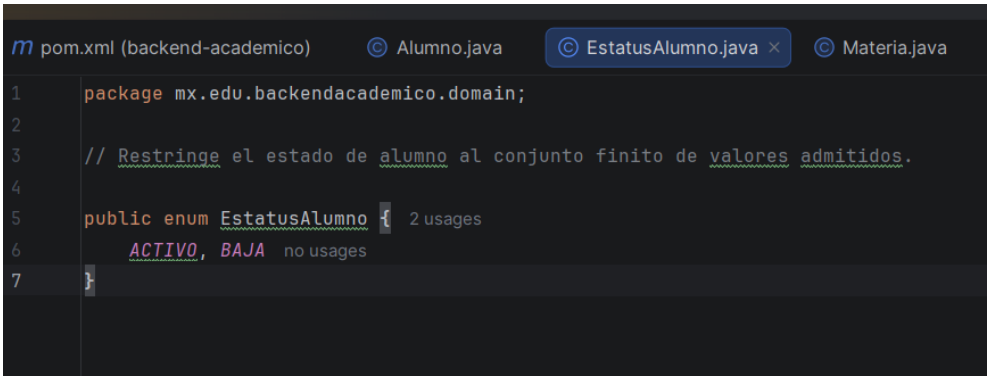
Pom.xml con revisión de Maven



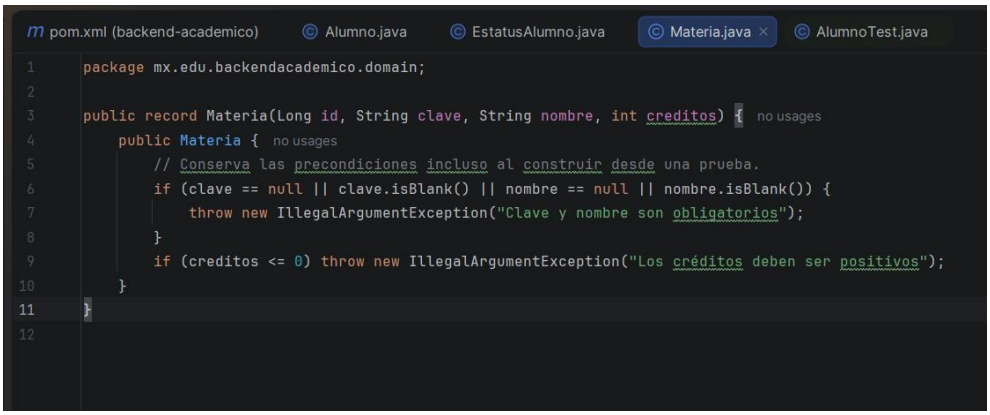
Alumno.java



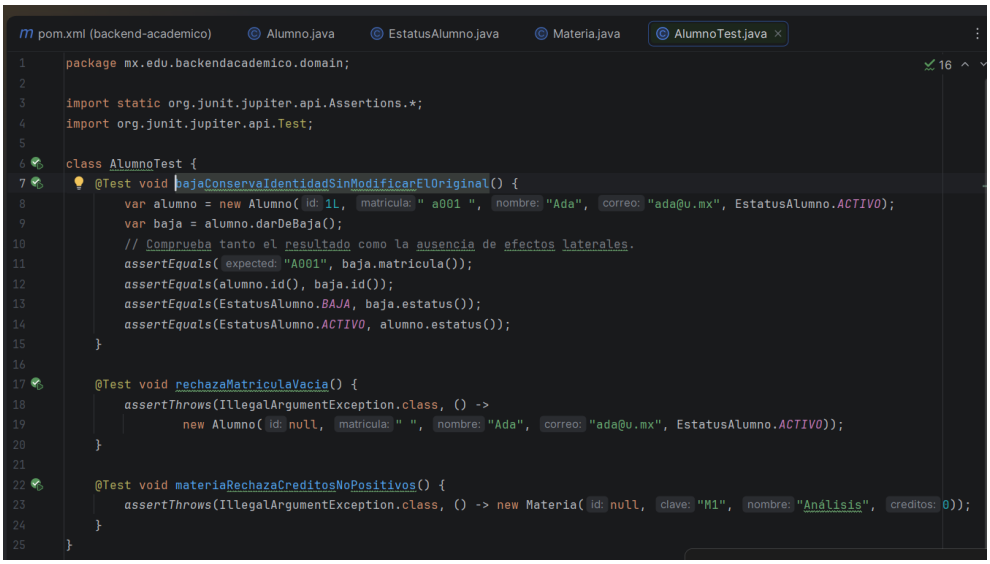
EstatusAlumno



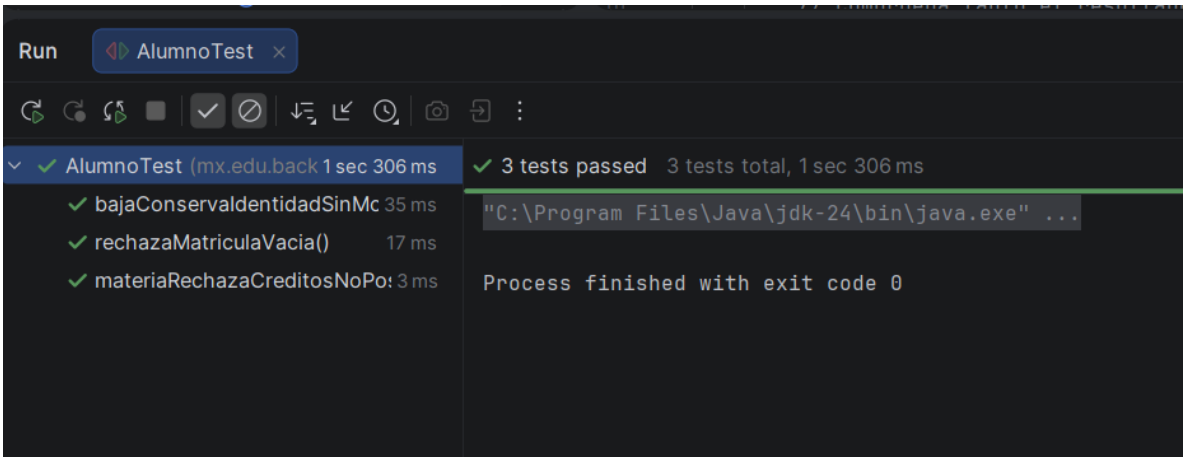
Materia

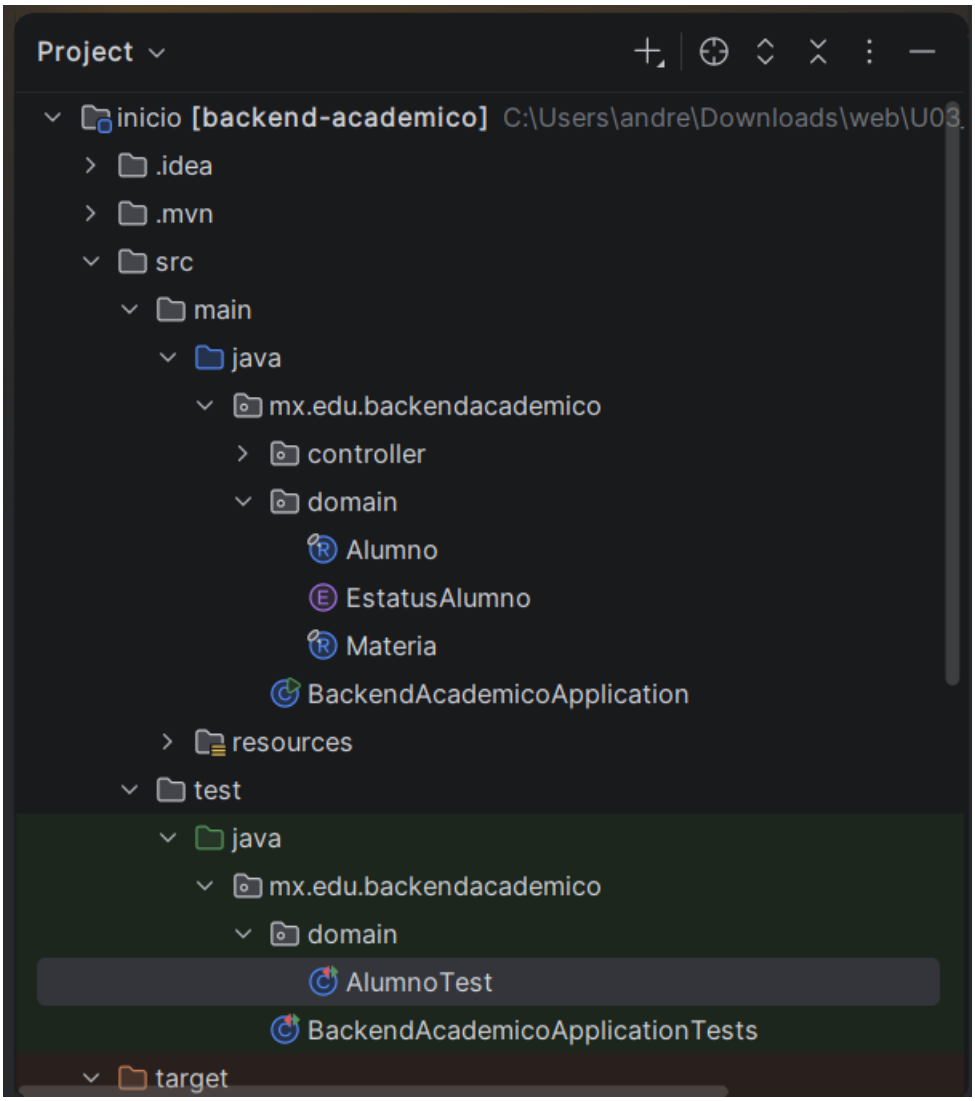


AlumnoTest



Pruebas:


Run
AlumnoTest
AlumnoTest (mx.edu.back 1 sec 306 ms
bajaConservaldentidadSinMc 35 ms
rechazaMatriculaVacia() 17 ms
materiaRechazaCreditosNoPo: 3 ms
3 tests passed 3 tests total, 1 sec 306 ms
"C:\Program Files\Java\jdk-24\bin\java.exe" ...
Process finished with exit code 0


Project
inicio [backend-academico] C:\Users\andre\Downloads\web\U03
.idea
.mvn
src
main
java
mx.edu.backendacademico
controller
domain
Alumno
EstatusAlumno
Materia
BackendAcademicoApplication
resources
test
java
mx.edu.backendacademico
domain
AlumnoTest
BackendAcademicoApplicationTests
target

```
PS C:\Users\andre\Downloads\web\U03_inicio\U03\inicio> .\mvnw.cmd -
Dtest=AlumnoTest test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice
-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u03 0.0.3-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 5 source files with javac [debug parameters release 21] to
target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-
academico ---
[INFO] skip non existing resourceDirectory
C:\Users\andre\Downloads\web\U03_inicio\U03\inicio\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico
---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug parameters release 21] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.097 s --
in mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.560 s
[INFO] Finished at: 2026-10-10T00:14:09-06:00
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\andre\Downloads\web\U03_inicio\U03\inicio> .\mvnw.cmd test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u03 0.0.3-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\web\U03_inicio\U03\inicio\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.BackendAcademicoApplicationTests
00:15:10.974 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect
default configuration classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]:
BackendAcademicoApplicationTests does not declare any static, non-private, non-final, nested classes annotated with
@Configuration.
00:15:11.132 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests
00:15:11.281 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect
default configuration classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]:
BackendAcademicoApplicationTests does not declare any static, non-private, non-final, nested classes annotated with
@Configuration.
00:15:11.282 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests

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 :: Spring Boot ::                (v4.1.0)

2026-10-10T00:15:11.770-06:00  INFO 20988 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: Starting BackendAcademicoApplicationTests using Java 24.0.2 with PID 20988 (started by andre in
C:\Users\andre\Downloads\web\U03_inicio\U03\inicio)
2026-10-10T00:15:11.772-06:00  INFO 20988 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: No active profile set, falling back to 1 default profile: "default"
2026-10-10T00:15:13.056-06:00  INFO 20988 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: Started BackendAcademicoApplicationTests in 1.662 seconds (process running for 3.1)
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the
JDK. Please add Mockito as an agent to your build as described in Mockito's documentation:
https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\andre\.m2\repository\net\bytebuddy\byte-buddy-
agent\1.18.10\byte-buddy-agent-1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath
has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.262 s -- in
mx.edu.backendacademico.BackendAcademicoApplicationTests
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.021 s -- in
mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.157 s
[INFO] Finished at: 2026-10-10T00:15:14-06:00
[INFO] ------------------------------------------------------------------------
```

COMPROBAR Y RAZONAR

Autoevaluación razonada

Tipos, invariantes e inmutabilidad · Semanas 3 · Sección 13 de 14 · JDK 25 · Java 21 · Boot 4.1.0

Imprimir unidad

¿Un record que contiene `ArrayList` es profundamente inmutable? No. Debes copiar y controlar sus elementos. **¿Una prueba del dominio necesita `@SpringBootTest`?** Sólo si la conducta depende de infraestructura Spring; estas reglas pueden probarse directamente.

PAUSA DE RAZONAMIENTO

Comprueba tu interpretación

Elige una respuesta y revisa el fundamento. Esta actividad no ejecuta las pruebas del proyecto.

¿Qué efecto tiene darDeBaja en el Alumno implementado?

- ○ Modifica los componentes del mismo record.
- ○ Vacía el repositorio de alumnos.
- ◉ Construye un valor nuevo y conserva el original.

Revisar respuesta

Interpretación correcta. La transición de este dominio se expresa como creación de un nuevo valor válido.

▶ Mis notas de esta sección

Reto

```java

    @Test
    void reactivarConservaElOriginal() {
        var alumno = new Alumno( id: 1L,  matricula: "A001",  nombre: "Ada",  correo: "ada@u.mx", EstatusAlumno.BAJA);
        var activo = alumno.reactivar();

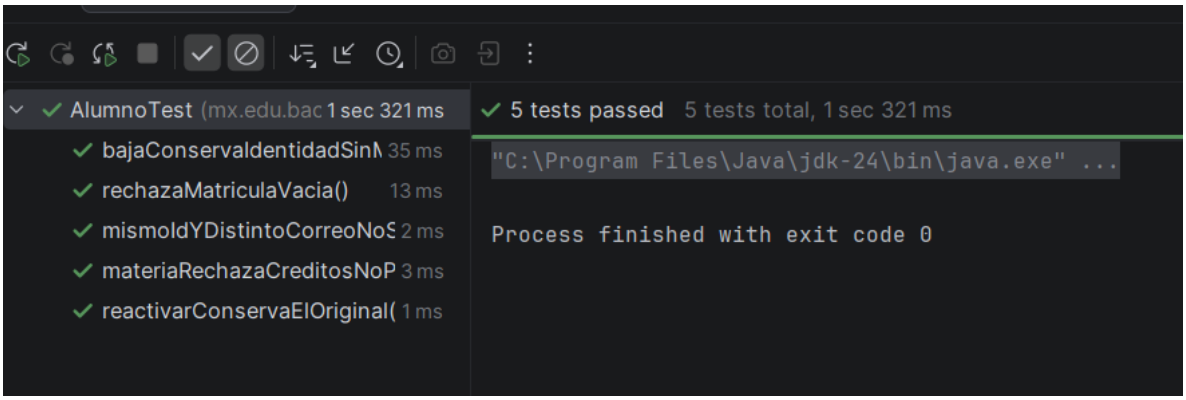
        assertEquals(EstatusAlumno.ACTIVO, activo.estatus());
        assertEquals(EstatusAlumno.BAJA, alumno.estatus());
        assertEquals(alumno.id(), activo.id());
        assertEquals(alumno.matricula(), activo.matricula());
    }


    @Test
    void mismoIdYDistintoCorreoNoSonIgualesPorValor() {
        var alumno1 = new Alumno( id: 1L,  matricula: "A001",  nombre: "Ada",  correo: "ada@u.mx", EstatusAlumno.ACTIVO);
        var alumno2 = new Alumno( id: 1L,  matricula: "A001",  nombre: "Ada",  correo: "ada@otro.mx", EstatusAlumno.ACTIVO);

        assertNotEquals(alumno1, alumno2);
        assertEquals(alumno1.id(), alumno2.id());
    }
```

```java
    }


    public Alumno reactivar() {  no usages
        return new Alumno(id, matricula, nombre, correo, EstatusAlumno.ACTIVO);
    }
}
```



```
PS C:\Users\andre\Downloads\web\U03_reto\U03\inicio> .\mvnw.cmd -Dtest=AlumnoTest
test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by
com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice
-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class
com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u03 0.0.3-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 5 source files with javac [debug parameters release 21] to
target\classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-
academico ---
[INFO] skip non existing resourceDirectory
C:\Users\andre\Downloads\web\U03_reto\U03\inicio\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico
---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug parameters release 21] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.130 s --
in mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.636 s
[INFO] Finished at: 2026-10-10T00:25:52-06:00
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\andre\Downloads\web\U03_reto\U03\inicio> .\mvnw.cmd test
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner
(file:/C:/Users/andre/.m2/wrapper/dists/apache-maven-
3.9.11/03d7e36a140982eea48e22c1dcac01d8862b2550b2939e09a0809bbc5182a5bc/lib/guice-5.1.0-classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< mx.edu:backend-academico >----------------------
[INFO] Building backend-academico-u03 0.0.3-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ backend-academico ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ backend-academico ---
[INFO] skip non existing resourceDirectory C:\Users\andre\Downloads\web\U03_reto\U03\inicio\src\test\resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ backend-academico ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ backend-academico ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running mx.edu.backendacademico.BackendAcademicoApplicationTests
00:27:29.717 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect
default configuration classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]:
BackendAcademicoApplicationTests does not declare any static, non-private, non-final, nested classes annotated with
@Configuration.
00:27:29.860 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests
00:27:29.980 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect
default configuration classes for test class [mx.edu.backendacademico.BackendAcademicoApplicationTests]:
BackendAcademicoApplicationTests does not declare any static, non-private, non-final, nested classes annotated with
@Configuration.
00:27:29.981 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found
@SpringBootConfiguration mx.edu.backendacademico.BackendAcademicoApplication for test class
mx.edu.backendacademico.BackendAcademicoApplicationTests

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 :: Spring Boot ::                (v4.1.0)

2026-10-10T00:27:30.433-06:00  INFO 34692 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: Starting BackendAcademicoApplicationTests using Java 24.0.2 with PID 34692 (started by andre in
C:\Users\andre\Downloads\web\U03_reto\U03\inicio)
2026-10-10T00:27:30.437-06:00  INFO 34692 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: No active profile set, falling back to 1 default profile: "default"
2026-10-10T00:27:31.596-06:00  INFO 34692 --- [backend-academico] [           main] m.e.b.BackendAcademicoApplicationTests
: Started BackendAcademicoApplicationTests in 1.526 seconds (process running for 2.874)
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the
JDK. Please add Mockito as an agent to your build as described in Mockito's documentation:
https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
WARNING: A Java agent has been loaded dynamically (C:\Users\andre\.m2\repository\net\bytebuddy\byte-buddy-
agent\1.18.10\byte-buddy-agent-1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath
has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.845 s -- in
mx.edu.backendacademico.BackendAcademicoApplicationTests
[INFO] Running mx.edu.backendacademico.domain.AlumnoTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.023 s -- in
mx.edu.backendacademico.domain.AlumnoTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.720 s
[INFO] Finished at: 2026-10-10T00:27:32-06:00
[INFO] ------------------------------------------------------------------------
```